
Problem Set 3

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1 Representations and Basis States

Even though our space is infinite-dimensional, we can isolate v_1, v_2 to consider a 2-dimensional subspace. The most general possible choice in this subspace, as illustrated by equation (1.1), is

$$|1\rangle = \cos \theta e^{i\phi_1} v_1 + \sin \theta e^{i\phi_2} v_2,$$

$$|2\rangle = \sin \theta e^{i\phi_3} v_1 - \cos \theta e^{i(\phi_3 + \phi_1 - \phi_2)} v_2.$$

Idfk what to do for b

2 Operators